

UNIT ROOT TESTS WHEN THE DATA ARE A TRIGONOMETRIC TRANSFORMATION OF AN INTEGRATED PROCESS

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Summary: This study establishes analytically what the asymptotic behavior of the Dickey-Fuller coefficient tests and the Dickey-Fuller t -statistic tests will be when the true data-generating process is a trigonometric function of an integrated process. Using some recently established limit theorems, it is shown that for such a data generating process, the asymptotic behavior of these unit root tests is reminiscent of that of a stationary series. These results confirm analytically the conclusions of Granger and Hallman in an earlier paper. The results of this paper are applied to the Fourier flexible form methodology.

1. Introduction

This paper establishes analytically what the asymptotic behavior of the Dickey-Fuller unit root tests will be when the true data-generating process is a trigonometric function of an integrated process. For example, the data could be generated as $\sin(x_t)$, where x_t is an integrated series. This problem has been analyzed mainly through simulations in Granger and Hallman (1991), and this paper gives the mathematical underpinning for their conclusions. Another paper that is related is Ermini and Granger (1993); in that paper, various moments and covariances are calculated that involve functions of integrated processes. Ermini and Granger's (1993) results are obtained by strongly relying on a normality assumption.

One important tool for the analysis of this paper is provided in de Jong (2001). In that paper, it is established that for functions $T(\cdot)$ that are periodic on $[c_1, c_2]$, for an integrated process x_t that

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satisfies some regularity conditions,

$$n^{-1/2} \sum_{t=1}^n (T(x_t) - (c_2 - c_1)^{-1} \int_{c_1}^{c_2} T(x) dx) \xrightarrow{d} N(0, \sigma^2), \quad (1)$$

where $\sigma^2 = E(\Delta x_t)^2$, Δx_t is i.i.d., and “ $\xrightarrow{d}$ ” denotes convergence in distribution. This paper extends the tools developed in de Jong (2001) somewhat in order to arrive at a complete asymptotic analysis of the problem under consideration. We then apply the analytical results of our paper to the Fourier Flexible Form methodology as it was used by Fisher, Fleissig and Serletis (2001).

2. Assumption and main results

We consider a time series x_t generated by

$$x_t = x_{t-1} + \varepsilon_t \quad (2)$$

where ε_t is a sequence of independent and identical distributed random variables with a continuous distribution, a mean of zero, and a variance of population disturbance σ_t^2 is defined as $\sigma^2 = T^{-1} \sum_{t=1}^T \varepsilon_t^2$. $\mathcal{F}_t = \sigma(\varepsilon_t, \varepsilon_{t-1}, \dots, \varepsilon_1, x_0)$ is the sigma field that includes all the information in ε_t until time period t , and x_0 is arbitrary. $y_t = T(x_t)$ is a trigonometric transformation of integrated process x_t that is allowed to be within one of two functions: sine and cosine. Below, let $\hat{\rho}$ denote the regression coefficient resulting from a regression of y_t on y_{t-1} , and let $\hat{\rho}_\mu$ denote the regression coefficient resulting from a regression of y_t on y_{t-1} and a constant. In all results below, we allow for $y_t = \sin(ax_t + b)$ for arbitrary constants a and b , and since $\sin(x) = \cos(\pi - x)$, our results therefore also cover the case of the cosine function.

For the convergence behavior of $\hat{\rho}$ and $\hat{\rho}_\mu$, the following result can be established:

Theorem 1 For the process x_t as defined before, for $y_t = \sin(ax_t + b)$, $a \in \mathbb{R} \setminus \{0\}$, $b \in \mathbb{R}$,

$$\hat{\rho} \xrightarrow{p} E \cos(a\varepsilon_t) \quad \text{and} \quad \hat{\rho}_\mu \xrightarrow{p} E \cos(a\varepsilon_t). \quad (3)$$

In the theorem above and elsewhere in this paper, “ $\xrightarrow{p}$ ” denotes convergence in probability. All proofs for this paper are deferred to the Mathematical Appendix.

For the regression coefficients $\hat{\rho}$ and $\hat{\rho}_\mu$, the following theorem establishes root- n consistency and asymptotic normality under the additional assumption that the distribution of ε_t is symmetric:

Theorem 2 For the processes x_t defined before, if ε_t has a symmetric distribution, for $y_t = \sin(ax_t + b)$, $a \in \mathbb{R} \setminus \{0\}$, $b \in \mathbb{R}$,

$$n^{1/2}(\hat{\rho} - E \cos(a\varepsilon_t)) \xrightarrow{d} N(0, V) \quad \text{and} \quad n^{1/2}(\hat{\rho}_\mu - E \cos(a\varepsilon_t)) \xrightarrow{d} N(0, V), \quad (4)$$

where

$$V = (3/8)E(\cos(a\varepsilon_t) - E \cos(a\varepsilon_t))^2 + (1/8)E(\sin(a\varepsilon_t))^2. \quad (5)$$

The above theorem implies that the Dickey-Fuller coefficient tests will go off to $-\infty$ at rate n - the same rate as would apply for stationary processes y_t - and therefore the Dickey-Fuller coefficient tests will asymptotically indicate stationarity.

Finally, we establish the asymptotic behavior of the Dickey-Fuller t -statistics $\hat{t}$ and $\hat{t}_\mu$ for the coefficients of y_{t-1} resulting respectively from a regression of Δy_t on y_{t-1} and a regression of Δy_t on y_{t-1} and a constant:

Theorem 3 For the processes x_t defined before, for $y_t = \sin(ax_t + b)$, $a \in \mathbb{R} \setminus \{0\}$, $b \in \mathbb{R}$, defining

$$c = (E \cos(a\varepsilon_t) - 1)(1 - (E \cos(a\varepsilon_t))^2)^{-1/2}, \quad (6)$$

we have

$$n^{-1/2}\hat{t} \xrightarrow{P} c \quad \text{and} \quad n^{-1/2}\hat{t}_\mu \xrightarrow{P} c. \quad (7)$$

Unlike Theorem 2, the result of Theorem 3 does not rely on a symmetry assumption for the distribution of ε_t . Unfortunately, deriving a full limit distribution result for $n^{-1/2}\hat{t} - c$ or $n^{-1/2}\hat{t}_\mu - c$, or relaxing the symmetry assumption, turned out to be complicated.

From our results, it is clear that the asymptotic behavior of the Dickey-Fuller coefficient and t -tests in terms of convergence rates is identical to that of the case of stationary random variables, and that the t -test will asymptotically indicate stationarity. This conclusion was also obtained through simulation in Granger and Hallman (1991).

3. Application: Fourier flexible form

Estimating indirect demand and cost function is an important topic in productivity and efficiency analysis. The Fourier flexible form was introduced by Gallant (1981) as a means of approximating unknown functional forms in a nonparametric or semiparametric way. Some recent empirical research that uses the Fourier flexible form methodology is Drake and Fleissig (2008), Feng and Serletis (2008, 2009), and Liu (2009). In this section, we focus on the analysis in Fisher et al. (2001), and explore how our results can shed light on the issues that arise in that paper.

In Fisher et al. (2001), the use of the Fourier Flexible Form implies that variables such as the consumer price index (CPI) are transformed by trigonometric functions. However, as noted in Fisher et al. (2001) and earlier in e.g. Ng (1995), the CPI series will have a unit root. Therefore, the Fourier Flexible Form methodology as applied in Fisher et al. (2001) depends on the use of trigonometric functions of unit root processes. These authors note that "As Granger (1995) points out, non-linear modelling of non-stationary variables is a new, complicated, and largely undeveloped area." This section uses the analytical development of the previous sections to revisit the results of Fisher et al. (2001) and see how our theoretical results translate to an empirical setting.

Table 1 reports the result of t_μ -statistic for the original and transformed series. The data that were used to obtain these results can be found at <http://econ.queensu.ca/jae/2001-v16.1>². From

²We used the sixth and last data series in this file. The CPI series is given the name "DP3" at the bottom of the data file and the name "P3" in the accompanying "readme" file. Results were calculated using the Gretl software package.

Table 1: t_μ -test for transformed CPI data; y_t denotes the CPI series

series	$n^{-1/2}t_\mu$	$\hat{c}$
y_t	11.630	
$\sin(y_t)$	0.529	-1.463
$\sin(2y_t)$	-0.061	-1.893
$\sin(3y_t)$	0.054	-1.949
$\cos(y_t)$	-0.149	-1.463
$\cos(2y_t)$	0.014	-1.893
$\cos(3y_t)$	-0.092	-1.949

The 5% percent critical value of Dickey-Fuller t -statistic is -2.86 . The $\hat{c}$ values were calculated by noting that

$$(n-1)^{-1} \sum_{t=2}^n \cos(j\Delta y_t) = -0.18824, -0.31858 \text{ and } -0.23579 \text{ for } j = 1, 2, 3, \text{ while}$$

$$(n-1)^{-1} \sum_{t=2}^n (\cos(j\Delta y_t))^2 = 0.34071, 0.51485 \text{ and } 0.59797 \text{ for } j = 1, 2, 3. .$$

this table, it seems clear that a great distance continues to exist between $n^{-1/2}t_\mu$ and $\hat{c}$. Therefore, it seems safe to conclude that one of our assumptions on x_t must fail in this setting. One possibility is that this distance is caused by the lack of independence of ε_t . A second possibility is that the asymptotic theory developed here needs substantially larger sample sizes and/or different values for $E\varepsilon_t^2$ to be valid. Finally, there is the possibility that the unit root model for CPI need refinement. However, the conclusion that can be draw is that an asymptotic theory for the estimators as developed in Fisher et al. (2001) based on the assumption that CPI is a random walk, if it can be developed at all, will likely not be informative about the small-sample properties of the estimators involved, since such a theory will be based on approximations between objects such as $n^{-1/2}t_\mu$ and $\hat{c}$.

4. Conclusion

In this paper we established the asymptotic behavior of the Dickey-Fuller coefficient tests and the Dickey-Fuller t -statistic tests when the true data-generating process is a trigonometric functions of an integrated process. These results can be used to explore the properties of Fourier flexible form estimators with nonstationary data in future research.

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Mathematical Appendix

Proof of Theorem 1:

By the law of large numbers for bounded martingale difference sequences,

$$n^{-1} \sum_{t=2}^n (\sin(ax_t + b) \sin(ax_{t-1} + b) - E(\sin(ax_t + b) \sin(ax_{t-1} + b) | \mathcal{F}_{t-1})) \xrightarrow{P} 0, \quad (8)$$

and therefore in order to find the probability limit of $(n-1)^{-1} \sum_{t=2}^n y_t y_{t-1}$, it suffices to consider

$$\begin{aligned} & n^{-1} \sum_{t=2}^n E(\sin(ax_t + b) \sin(ax_{t-1} + b) | \mathcal{F}_{t-1}) \\ &= n^{-1} \sum_{t=2}^n (\sin(ax_{t-1} + b))^2 E \cos(a\varepsilon_t) + \sin(ax_{t-1} + b) \cos(ax_{t-1} + b) E \sin(a\varepsilon_t) \\ &= E \cos(a\varepsilon_t) n^{-1} \sum_{t=2}^n \sin(ax_{t-1} + b)^2 + E \sin(a\varepsilon_t) n^{-1} \sum_{t=2}^n \sin(ax_{t-1} + b) \cos(ax_{t-1} + b). \end{aligned} \quad (9)$$

From Theorem 2 of de Jong (2001), i.e. the result of Equation (1), it follows that

$$n^{-1} \sum_{t=2}^n \sin(ax_{t-1} + b)^2 \xrightarrow{P} 1/2 \quad \text{and} \quad n^{-1} \sum_{t=2}^n \sin(ax_{t-1} + b) \cos(ax_{t-1} + b) \xrightarrow{P} 0, \quad (10)$$

and therefore

$$n^{-1} \sum_{t=2}^n E(\sin(ax_t + b) \sin(ax_{t-1} + b) | \mathcal{F}_{t-1}) \xrightarrow{p} (1/2) E \cos(a\varepsilon_t). \quad (11)$$

It now follows that

$$\hat{\rho} = \frac{n^{-1} \sum_{t=2}^n y_t y_{t-1}}{n^{-1} \sum_{t=2}^n y_{t-1}^2} \xrightarrow{p} \frac{(1/2) E \cos(a\varepsilon_t)}{(1/2)} = E \cos(a\varepsilon_t). \quad (12)$$

For $\hat{\rho}_\mu$, the same result follows by noting that

$$\hat{\rho}_\mu = \frac{n^{-1} \sum_{t=2}^n (y_t - \bar{y})(y_{t-1} - \bar{y})}{n^{-1} \sum_{t=2}^n (y_{t-1} - \bar{y})^2}, \quad (13)$$

and by noting that again by the result of Equation (1), $\bar{y} \xrightarrow{p} 0$. ■

Proof of Theorem 2:

For $y_t = \sin(a + bx_t)$, we have

$$\begin{aligned} n^{1/2}(\hat{\rho} - E \cos(a\varepsilon_t)) &= \frac{n^{-1/2} \sum_{t=2}^n (\sin(ax_t + b) \sin(ax_{t-1} + b) - \sin^2(ax_{t-1} + b) E \cos(a\varepsilon_t))}{n^{-1} \sum_{t=2}^n \sin^2(ax_{t-1} + b)} \\ &= \frac{n^{-1/2} \sum_{t=2}^n (\sin(ax_{t-1} + b) \cos(ax_{t-1} + b) \sin(a\varepsilon_t) + \sin(ax_{t-1} + b)^2 \cos(a\varepsilon_t) - \sin(ax_{t-1} + b)^2 E \cos(a\varepsilon_t))}{n^{-1} \sum_{t=2}^n \sin^2(ax_{t-1} + b)}. \end{aligned} \quad (14)$$

Now, noting that the denominator converges in probability to 1/2 as before, and in addition note that the summands g_t in the numerator are martingale differences with respect to $\mathcal{F}_t$ by symmetry of the distribution of ε_t , which implies that $E \sin(\varepsilon_t) = 0$. We now apply the martingale difference central limit theorem; see e.g. Theorem 3.2 of Hall and Heyde (1980). To verify the conditions of this theorem, it now only remains to be shown that $n^{-1} \sum_{t=2}^n g_t^2 \xrightarrow{p} V \in (0, \infty)$. This will be true because

$$\begin{aligned} n^{-1} \sum_{t=2}^n g_t^2 &= n^{-1} \sum_{t=2}^n \sin(ax_{t-1} + b)^2 (\sin(ax_{t-1} + b) (\cos(a\varepsilon_t) - E \cos(a\varepsilon_t)) + \cos(ax_{t-1} + b) \sin(a\varepsilon_t))^2 \\ &= n^{-1} \sum_{t=2}^n \sin(ax_{t-1} + b)^4 (\cos(a\varepsilon_t) - E \cos(a\varepsilon_t))^2 \\ &\quad + n^{-1} \sum_{t=2}^n \sin(ax_{t-1} + b)^2 \cos(ax_{t-1} + b)^2 \sin(a\varepsilon_t)^2 \\ &\quad + n^{-1} \sum_{t=2}^n 2 \sin(ax_{t-1} + b)^3 \cos(ax_{t-1} + b) (\cos(a\varepsilon_t) - E \cos(a\varepsilon_t)) \sin(a\varepsilon_t). \end{aligned} \quad (15)$$

By the martingale difference law of large numbers, the last expression equals

$$o_P(1) + n^{-1} \sum_{t=1}^n \sin(ax_{t-1} + b)^4 E (\cos(a\varepsilon_t) - E \cos(a\varepsilon_t))^2$$

$$\begin{aligned}
& + n^{-1} \sum_{t=1}^n \sin(ax_{t-1} + b)^2 \cos(ax_{t-1} + b)^2 E(\sin(a\varepsilon_t))^2 \\
& + n^{-1} \sum_{t=1}^n 2 \sin(ax_{t-1} + b)^3 \cos(ax_{t-1} + b) E((\cos(a\varepsilon_t) - E \cos(a\varepsilon_t)) \sin(a\varepsilon_t)).
\end{aligned} \tag{16}$$

By Theorem 2 of de Jong (2001) as quoted in Equation (1), we know that

$$n^{-1} \sum_{t=1}^n \sin(ax_{t-1} + b)^4 \xrightarrow{p} (2\pi)^{-1} \int_{-\pi}^{\pi} \sin^4(x) dx = 3/8, \tag{17}$$

$$n^{-1} \sum_{t=1}^n \sin(ax_{t-1} + b)^2 \cos(ax_{t-1} + b)^2 \xrightarrow{p} (2\pi)^{-1} \int_{-\pi}^{\pi} \sin^2(x) \cos^2(x) dx = 1/8, \tag{18}$$

and

$$n^{-1} \sum_{t=1}^n 2 \sin(ax_{t-1} + b)^3 \cos(ax_{t-1} + b) \xrightarrow{p} (2\pi)^{-1} \int_{-\pi}^{\pi} 2 \sin(x)^3 \cos(x) dx = 0. \tag{19}$$

Therefore, it follows that

$$n^{-1} \sum_{t=1}^n g_t^2 \xrightarrow{p} (3/8) E(\cos(a\varepsilon_t) - E \cos(a\varepsilon_t))^2 + (1/8) E(\sin(a\varepsilon_t))^2 = V. \tag{20}$$

For $\hat{\rho}_\mu$, note that

$$|n^{1/2}(\hat{\rho}_\mu - \hat{\rho})| \leq o_P(1) + 2n^{1/2}\bar{y}^2 = O_P(n^{-1/2}), \tag{21}$$

implying that the same limit as for $n^{1/2}(\hat{\rho} - E \cos(a\varepsilon_t))$ results for $n^{1/2}(\hat{\rho}_\mu - E \cos(a\varepsilon_t))$ as well, and this observation completes the proof of Theorem 2. ■

Proof of Theorem 3:

Using the results obtained in the proof of Theorem 1,

$$\begin{aligned}
s^2 &= (n-1)^{-1} \sum_{t=2}^n (y_t - \hat{\rho} y_{t-1})^2 = o_P(1) + n^{-1} \sum_{t=2}^n y_t^2 - 2\hat{\rho} n^{-1} \sum_{t=2}^n y_t y_{t-1} + \hat{\rho}^2 n^{-1} \sum_{t=2}^n y_{t-1}^2 \\
&\xrightarrow{p} (1/2) - 2E \cos(a\varepsilon_t) (1/2) E \cos(a\varepsilon_t) + (E \cos(a\varepsilon_t))^2 (1/2) \\
&= (1/2) - (1/2) (E \cos(a\varepsilon_t))^2.
\end{aligned} \tag{22}$$

Therefore, it now follows that

$$\begin{aligned}
n^{-1/2} \hat{t} &= n^{-1/2} \frac{(n-1)^{1/2} (\hat{\rho} - 1)}{(s^2 / ((n-1)^{-1} \sum_{t=2}^n y_{t-1}^2))^{1/2}} \\
&= n^{-1/2} \frac{(n-1)^{1/2} (\hat{\rho} - E \cos(a\varepsilon_t))}{(s^2 / ((n-1)^{-1} \sum_{t=2}^n y_{t-1}^2))^{1/2}} + n^{-1/2} \frac{(n-1)^{1/2} (E \cos(a\varepsilon_t) - 1)}{(s^2 / ((n-1)^{-1} \sum_{t=2}^n y_{t-1}^2))^{1/2}} \\
&\xrightarrow{p} (E \cos(a\varepsilon_t) - 1) (1 - (E \cos(a\varepsilon_t))^2)^{-1/2}.
\end{aligned} \tag{23}$$

For $n^{-1/2} \hat{t}_\mu$, the same result holds, because the $\bar{y}$ that would appear in the expression for $\hat{t}_\mu$ converges to 0 in probability, and therefore the difference between $n^{-1/2} \hat{t}$ and $n^{-1/2} \hat{t}_\mu$ converges to zero in probability asymptotically. ■

